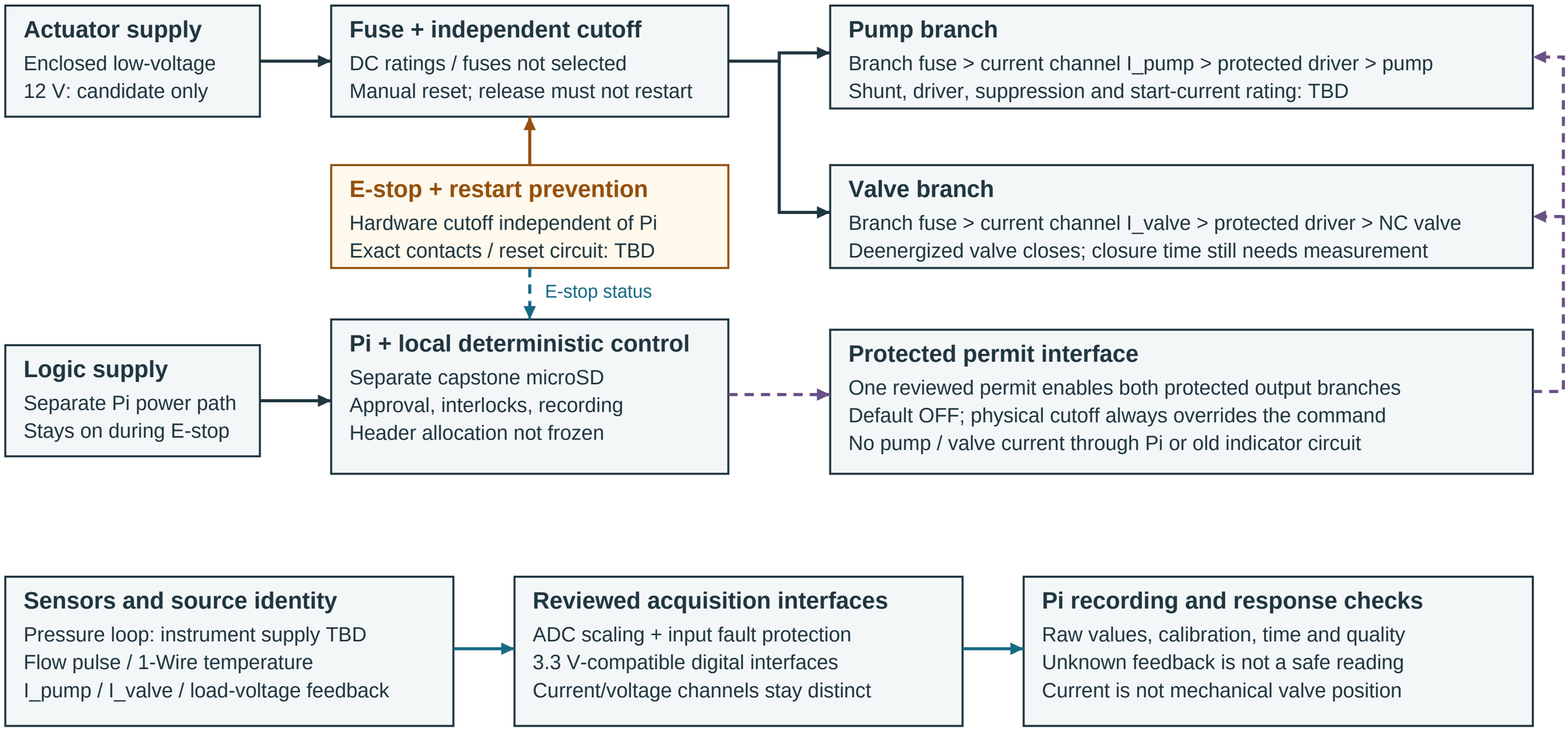
Electrical architecture for the proposed extension

Power, command and measured response are separate. Arrows show functional paths, not individual conductors.

R0 DESIGN REVIEW ONLY - no pin-level build release, ERC/SPICE sign-off or physical validation



CHECK FOUND: a 10 V-minimum sensor plus 150 ohm burden needs >=13 V at 20 mA, before cable losses.
The 150 ohm receiver is an unaccepted example: at 24 mA it develops 3.6 V, beyond a 3.3 V ADC rail.
Grounding/isolation, return conductors, GPIO pins, I2C addresses and component ratings need the final netlist.
Black = actuator / logic power; dashed purple = permit; blue = measurement. These are functional connections.

Hydraulic layout for the proposed water loop

Room-temperature water; open reservoir; low-head pump. Pressure target, trip and protection setting remain unfrozen.

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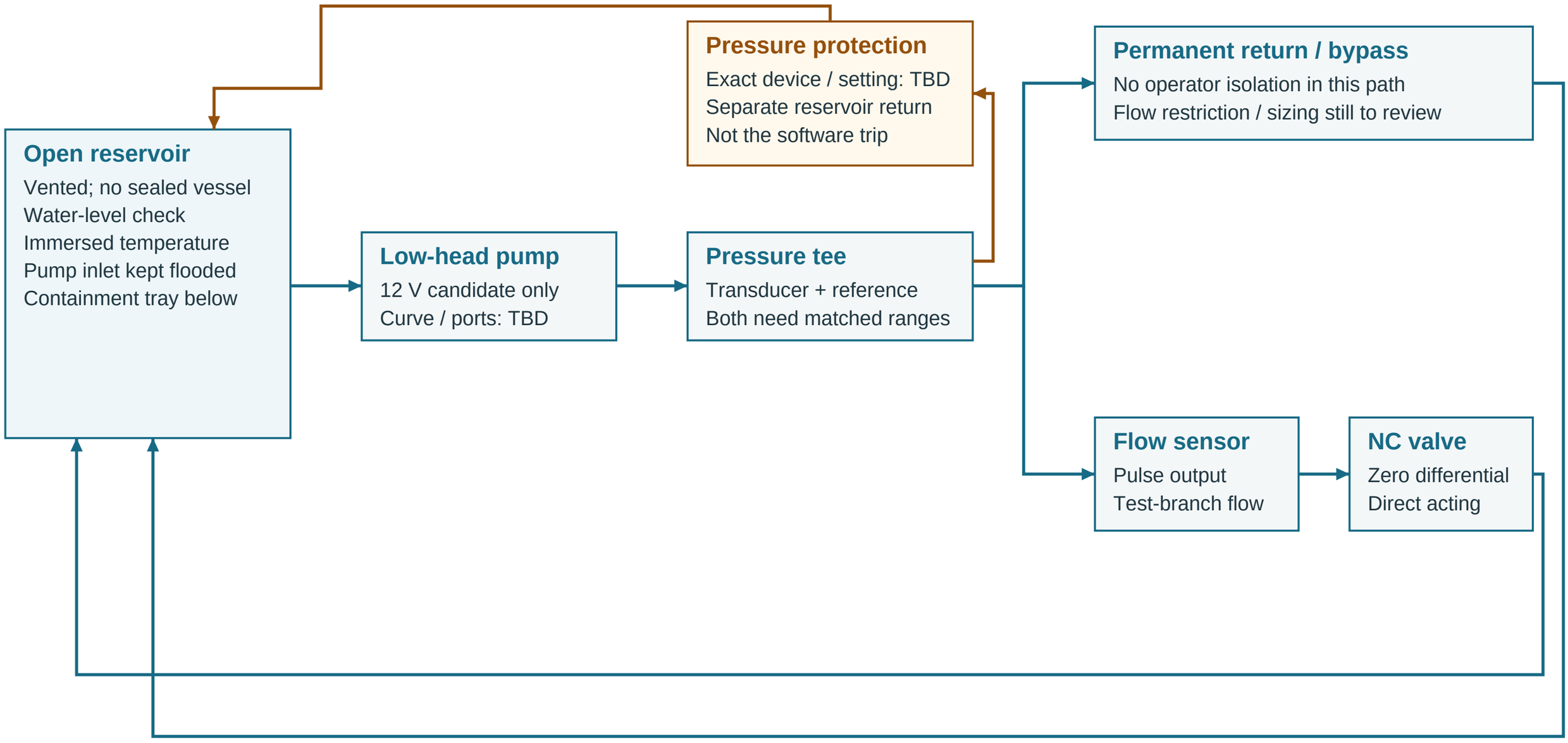

The bypass and protection returns are shown separately. No dead-head or blocked-return trial is authorized.
Flow measures the valve test branch. Pump current, valve current and flow provide different evidence.
Freeze operating pressure, software trip, protection tolerance and uncertainty below the weakest working rating.

Design review findings and required validation

Desk checks completed on stated assumptions. No new hardware has been energized, calibrated or accepted.

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Checked calculation: sensor supply compliance

Candidate sheet: sensor minimum = 10 V; output = 4-20 mA.
With R=150 ohm and I=20 mA: burden drop = 3.0 V.
12 V supply leaves 9.0 V at the sensor before cable losses.
Result: FAIL for this combination; instrumentation supply is TBD.

Checked calculation: ADC input margin

R=150 ohm: 4 mA -> 0.6 V; 20 mA -> 3.0 V.
Illustrative overrange: 22 mA -> 3.3 V; 24 mA -> 3.6 V.
A 3.3 V ADS1115 input must stay within its supply rails.
Result: receiver / overrange protection design remains open.

Component evidence still on HOLD

Pressure sheet label says 0-50 kPa; its table ends at 20 kPa.
Exact valve coil/zero-differential data and quote need acceptance.
Pump curve, fittings, reference and protection are not frozen.

Measured-feedback design revision

Use separate current measurements for pump and valve.
One combined reading cannot confirm each load independently.
Two channels are proposed; revised delivered basket must fit budget.

Before a construction circuit can be released

1. Review current bench photos, terminal identities and continuity against the canonical Phase 3 record.
2. Accept exact parts and datasheets; size supply, fuses, driver, shunts, suppression and DC cutoff together.
3. Complete the pin-by-pin netlist, grounding/isolations, input conditioning and wiring schedule.
4. Perform netlist/ERC and appropriate worst-case/transient calculations; record assumptions and results.
5. Unpowered inspection > dry feedback > passive leak checks > read-only sensing > calibration.
6. Attended protected trial > E-stop/no-restart checks > integrated cycles > 60-minute acquisition.

Source basis and claim boundary

TI ADS1115 Rev E: https://www.ti.com/lit/ds/symlink/ads1115.pdf
TI INA219 Rev G: https://www.ti.com/lit/ds/symlink/ina219.pdf
Candidate pressure sheet: https://robu.in/wp-content/uploads/2025/06/R222349-PR19C-PA-050K.pdf
Canonical Phase 3 SVG and current HARDWARE_REVIEW.md were inspected. They do not validate the extension.
No ERC/SPICE run, independent electrical review, physical test, final BOM or operating-pressure acceptance is claimed.